Variable Selection for Wasserstein Robust Hypothesis Testing

1 Problem formulation

Throughout this paper, we consider the variable selection problem for robust hypothesis testing. Let $[D] = \{1, \ldots, D\}$ and d be a positive integer denoting the sparsity budget. Given a detector $f : \mathbb{R}^D \to \mathbb{R}$, we follow the convention in [2] to define the worst-case risk as

$$R(f) = \sup_{\mathbb{P}_0 \in \mathcal{P}_0, \mathbb{P}_1 \in \mathcal{P}_1} \left\{ \mathbb{E}_{\mathbb{P}_0} \ell(-f(X)) + \mathbb{E}_{\mathbb{P}_1} \ell(f(X)) \right\}, \tag{1}$$

where $\mathcal{P}_k, k = 0, 1$ denote the Wasserstein distance-based ambiguity sets, and $\ell(\cdot)$ is the generating function that is nonnegative-valued, non-decreasing, convex, and satisfies $\ell(0) = 1, \lim_{t \to -\infty} \ell(t) = 0$. For $k = 0, 1$, assume we have n_k available observations in the group k, which are denoted as $\{x_i^{(k)}\}_{i=1}^{n_k}$. We denote the corresponding empirical distribution as $\widehat{\mathbb{P}}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} \delta_{x_i^{(k)}}$, and construct the Wasserstein ambiguity set as

$$\mathcal{P}_k = \left\{ \mathbb{P}_k : \mathcal{W}_1(\widehat{\mathbb{P}}_k, \mathbb{P}_k) \le \rho_k \right\}.$$

Here, $\mathcal{W}_1(\cdot, \cdot)$ refers to the 1-Wasserstein distance equipped with the ℓ_1-norm.

For $S \subseteq [D]$, define the collection of detectors whose input only depends on coordinates belonging to S:

$$\mathcal{F}_S = \left\{ f : \mathbb{R}^D \to \mathbb{R}, f(x) = g(x[S]) \text{ for some measurable function } g : \mathbb{R}^{|S|} \to \mathbb{R} \right\}.$$

Note that for a given vector $x \in \mathbb{R}^D$, we use $x[i]$ to denote its i-th coordinate, and $x[S]$ to denote the concatenation of its coordinates whose indices belong to S. Our proposed variable selection problem can be formulated as the following combinatorial optimization problem

$$\min_{S \subseteq [D], |S| \le d} \min_{f \in \mathcal{F}_S} R(f). \tag{2}$$

2 Convex formulation

Assume $n_0, n_1 \ge 1$, $\rho_0, \rho_1 > 0$, and that ℓ is finite on $\mathbb{R}$. All distributions have finite first moments. Fix $S \subseteq [D]$ and write

$$V(S) = \min_{f \in \mathcal{F}_S} R(f), \qquad \mathcal{X} = \{x_i^{(0)} : i \in [n_0]\} \cup \{x_j^{(1)} : j \in [n_1]\}.$$

The following lemma extends the finite-support formulation of [3, Lemma 2, equation (7)] to the generating loss in Section 1. Its proof, including projection and detector extension, is given in Appendix A.

Lemma 2.1 (Finite LFD reformulation)**.** *The fixed-selection risk is*

$$\begin{aligned}
V(S) = \max_{p, \gamma \ge 0} \quad & \sum_{y \in \mathcal{X}} \inf_{v \in \mathbb{R}} \{p_0(y)\ell(-v) + p_1(y)\ell(v)\} \\
\textit{subject to} \quad & \sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} = \frac{1}{n_k}, && i \in [n_k],\ k = 0, 1, \\
& p_k(y) = \sum_{i=1}^{n_k} \gamma_{i,y}^{(k)}, && y \in \mathcal{X},\ k = 0, 1, \\
& \sum_{i=1}^{n_k} \sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} \|x_i^{(k)}[S] - y[S]\|_1 \le \rho_k, && k = 0, 1.
\end{aligned} \tag{3}$$

Here $\gamma_{i,y}^{(k)}$ is the mass transported from $x_i^{(k)}[S]$ to $y[S]$, and $p_k(y)$ is the target mass.

Write g_y for the score assigned to $y[S]$. Substituting the target masses gives

$$\sum_{y\in\mathcal{X}} \inf_{v\in\mathbb{R}} \{p_0(y)\ell(-v) + p_1(y)\ell(v)\} = \inf_g \left\{ \sum_{i,y} \gamma_{i,y}^{(0)} \ell(-g_y) + \sum_{j,y} \gamma_{j,y}^{(1)} \ell(g_y) \right\}.$$

The objective is convex in g and linear in γ, and the transport set is compact and convex. The finite-dimensional minimax theorem gives

$$V(S) = \inf_g \left\{ \max_{\gamma^{(0)}} \sum_{i,y} \gamma_{i,y}^{(0)} \ell(-g_y) + \max_{\gamma^{(1)}} \sum_{j,y} \gamma_{j,y}^{(1)} \ell(g_y) \right\},$$

where each maximum satisfies its source-mass and transport-budget constraints in (3).

For a fixed score vector g, associate multipliers $a_i^{(0)} \in \mathbb{R}$ and $\lambda_0 \geq 0$ with the mass and transport constraints in (3), respectively. The Lagrangian of the maximization over $\gamma^{(0)}$ is

$$\begin{aligned} \mathcal{L}_0 = & \sum_{i,y} \gamma_{i,y}^{(0)} \ell(-g_y) + \sum_i a_i^{(0)} \left(\frac{1}{n_0} - \sum_y \gamma_{i,y}^{(0)} \right) \\ & + \lambda_0 \left(\rho_0 - \sum_{i,y} \gamma_{i,y}^{(0)} \|x_i^{(0)}[S] - y[S]\|_1 \right) \\ = & \frac{1}{n_0} \sum_i a_i^{(0)} + \rho_0 \lambda_0 \\ & + \sum_{i,y} \gamma_{i,y}^{(0)} \left[\ell(-g_y) - a_i^{(0)} - \lambda_0 \|x_i^{(0)}[S] - y[S]\|_1 \right]. \end{aligned} \tag{4}$$

Thus

$$\sup_{\gamma^{(0)}\geq 0} \mathcal{L}_0 < \infty \iff \ell(-g_y) \leq a_i^{(0)} + \lambda_0 \|x_i^{(0)}[S] - y[S]\|_1, \quad i \in [n_0],\ y \in \mathcal{X}.$$

Linear programming duality applied to both classes gives

$$\begin{aligned} V(S) = \min_{g,a,\lambda} \quad & \frac{1}{n_0} \sum_i a_i^{(0)} + \frac{1}{n_1} \sum_j a_j^{(1)} + \rho_0\lambda_0 + \rho_1\lambda_1 \\ \text{subject to} \quad & \ell(-g_y) \leq a_i^{(0)} + \lambda_0 \|x_i^{(0)}[S] - y[S]\|_1, \qquad i \in [n_0],\ y \in \mathcal{X}, \\ & \ell(g_y) \leq a_j^{(1)} + \lambda_1 \|x_j^{(1)}[S] - y[S]\|_1, \qquad j \in [n_1],\ y \in \mathcal{X}, \\ & g \in \mathbb{R}^{\mathcal{X}}, \quad a^{(k)} \in \mathbb{R}^{n_k}, \quad \lambda_k \geq 0, \qquad k = 0, 1. \end{aligned} \tag{5}$$

Denote this objective by $J(a, \lambda)$. The equivalence with the risk over measurable detectors is established in Appendix A.

Let $z = \mathbf{1}_S$ and $\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \leq d\}$. The transport cost decomposes directly into coordinates:

$$\|x_i^{(k)}[S] - y[S]\|_1 = \sum_{m=1}^{D} z_m |x_i^{(k)}[m] - y[m]|.$$

Introduce w_{km} to encode coordinate activation and extend the selection indicators to $z \in [0,1]^D$. Define

$$\begin{aligned} F(z) = \min_{g,a,\lambda,w} \quad & J(a, \lambda) \\ \text{subject to} \quad & \ell(-g_y) \leq a_i^{(0)} + \sum_m w_{0m} |x_i^{(0)}[m] - y[m]|, \quad i \in [n_0],\ y \in \mathcal{X}, \\ & \ell(g_y) \leq a_j^{(1)} + \sum_m w_{1m} |x_j^{(1)}[m] - y[m]|, \qquad j \in [n_1],\ y \in \mathcal{X}, \\ & 0 \leq w_{km} \leq \lambda_k, \qquad k = 0, 1,\ m \in [D], \\ & \rho_0 w_{0m} + \rho_1 w_{1m} \leq 2 z_m, \qquad m \in [D], \\ & g \in \mathbb{R}^{\mathcal{X}}, \quad a^{(k)} \in \mathbb{R}^{n_k}, \quad \lambda_k \geq 0, \qquad k = 0, 1. \end{aligned} \tag{6}$$

The constraints are jointly convex in (z, g, a, λ, w), so partial minimization defines a convex function F on $[0,1]^D$. At $z = \mathbf{1}_S$, the constraints impose $w_{km} = 0$ for $m \notin S$ and admit an optimum with $w_{km} = \lambda_k$ for $m \in S$. Consequently,

$$F(\mathbf{1}_S) = V(S), \qquad \min_{|S| \le d} V(S) = \min_{z \in \mathcal{Z}_d} F(z). \tag{7}$$

The continuous extension preserves the objective at every binary selection; minimizing it over the continuous budget set gives a relaxation. The discrete problem has the form of the logical-constraint models in [1, problems (2.1)–(2.2)]. The exactness and attainment arguments are included in Appendix B.

Associate multipliers $\gamma_{i,y}^{(k)} \ge 0$ with the loss constraints in (6) and $\mu_m \ge 0$ with the coordinate constraints. The resulting dual is as follows.

Proposition 2.2 (Dual representation)**.** *For every $z \in [0,1]^D$,*

$$F(z) = \max_{(\gamma,\mu) \in \mathcal{D}} \{H(\gamma) - 2\mu^\top z\}, \tag{8}$$

where

$$H(\gamma) = \sum_{y \in \mathcal{X}} \inf_{v \in \mathbb{R}} \left\{ \sum_{i=1}^{n_0} \gamma_{i,y}^{(0)} \ell(-v) + \sum_{j=1}^{n_1} \gamma_{j,y}^{(1)} \ell(v) \right\}, \tag{9}$$

and

$$\mathcal{D} = \left\{ (\gamma,\mu) \ge 0 : \begin{array}{ll} \displaystyle\sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} = 1/n_k, & i \in [n_k],\ k = 0,1, \\ \displaystyle\sum_{m=1}^{D} \left(\sum_{i=1}^{n_k} \sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} |x_i^{(k)}[m] - y[m]| - \rho_k \mu_m \right)_+ \le \rho_k, & k = 0,1 \end{array} \right\}, \tag{10}$$

with $(v)_+ = \max\{v, 0\}$. The maximum is attained. For an optimizer (γ^t, μ^t) at z^t,

$$\begin{aligned} F(z) &\ge c_t(z) := H(\gamma^t) - 2(\mu^t)^\top z \\ &= F(z^t) - 2(\mu^t)^\top (z - z^t), \qquad z \in [0,1]^D, \end{aligned} \tag{11}$$

and $c_t(z^t) = F(z^t)$.

The proof is given in Appendix B.

3 Outer-approximation approach

Proposition 2.2 yields an outer-approximation scheme of the form in [1, Theorem 2.5 and Algorithm 3.1]. Let $\mathrm{OPT} = \min_{z \in \mathcal{Z}_d} F(z)$. After evaluating $z^1, \dots, z^t$ and obtaining their dual optimizers, let $U_t = \min_{s \le t} F(z^s)$ and solve

$$\begin{aligned} (z^{t+1}, L_t) \in \operatorname*{arg\,min}_{z \in \mathcal{Z}_d,\ \eta \ge 0} \quad & \eta \\ \text{subject to} \quad & \eta \ge F(z^s) - 2(\mu^s)^\top (z - z^s), \qquad s = 1, \dots, t. \end{aligned} \tag{12}$$

By (11),

$$L_t \le \mathrm{OPT} \le U_t. \tag{13}$$

Starting from any $z^1 \in \mathcal{Z}_d$, alternate the convex subproblem (6)–(8) and the binary linear master (12). Stop when $U_t - L_t \le \varepsilon$ and return the best evaluated selection. Appendix A recovers its detector, with risk U_t and optimality gap at most ε. For an inexact inner solve, use the actual feasible dual expression $H(\gamma^t) - 2(\mu^t)^\top z$ as the cut and a primal feasible objective as the upper bound; the master must provide a global lower bound.

A Proof of Lemma 2.1

Proof. Maximization over γ and minimization over (g, a, λ) below are subject to (3) and (5), respectively. Let $S^c = [D] \setminus S$. For each source–target pair, define the lifted target by

$$\widetilde{y}_i^{(k)}[S] = y[S], \qquad \widetilde{y}_i^{(k)}[S^c] = x_i^{(k)}[S^c], \qquad \mathbb{P}_k^{\gamma} = \sum_{i,y} \gamma_{i,y}^{(k)} \delta_{\widetilde{y}_i^{(k)}}.$$

Then

$$\mathcal{W}_1(\widehat{\mathbb{P}}_k, \mathbb{P}_k^{\gamma}) \le \sum_{i,y} \gamma_{i,y}^{(k)} \|x_i^{(k)} - \widetilde{y}_i^{(k)}\|_1 = \sum_{i,y} \gamma_{i,y}^{(k)} \|x_i^{(k)}[S] - y[S]\|_1 \le \rho_k.$$

Consequently, for every $f(x) = g(x[S])$,

$$\begin{aligned} R(f) &\ge \sum_y \{p_0(y)\ell(-g(y[S])) + p_1(y)\ell(g(y[S]))\} \\ &\ge H(\gamma), \qquad \max_{\gamma} H(\gamma) \le V(S), \end{aligned} \tag{14}$$

where H is the objective in (9). Finite-dimensional minimax and the linear programming duality in (4)–(5) give

$$\begin{aligned} \max_{\gamma} H(\gamma) &= \inf_{g \in \mathbb{R}^{\mathcal{X}}} \max_{\gamma} \sum_y \{p_0(y)\ell(-g_y) + p_1(y)\ell(g_y)\} \\ &= \inf_{(g,a,\lambda)} J(a, \lambda). \end{aligned} \tag{15}$$

Now fix any feasible (g, a, λ). Coincident projected targets have identical constraint columns, so their scores may be chosen equal. Set $g(y[S]) = g_y$ and define

$$E_k(\xi) = \min_{r \in [n_k]} \{a_r^{(k)} + \lambda_k \|x_r^{(k)}[S] - \xi\|_1\}, \qquad k = 0, 1.$$

Outside the projected target set, choose the smallest minimizing indices i, j for $E_0(\xi), E_1(\xi)$, respectively, and put

$$r_0 = \|x_i^{(0)}[S] - \xi\|_1, \quad r_1 = \|x_j^{(1)}[S] - \xi\|_1, \quad b = \|x_i^{(0)}[S] - x_j^{(1)}[S]\|_1,$$

$$\tau = \begin{cases} \min\{1, r_0/b\}, & b > 0, \\ 0, & b = 0, \end{cases} \qquad g(\xi) = (1-\tau) g_{x_i^{(0)}} + \tau g_{x_j^{(1)}}. \tag{16}$$

The triangle inequality gives

$$b \le r_0 + r_1, \qquad \tau b \le r_0, \qquad (1-\tau) b \le r_1.$$

Hence convexity and the source and cross-target constraints imply

$$\begin{aligned} \ell(-g(\xi)) &\le (1-\tau)\ell(-g_{x_i^{(0)}}) + \tau \ell(-g_{x_j^{(1)}}) \\ &\le a_i^{(0)} + \lambda_0 \tau b \le a_i^{(0)} + \lambda_0 r_0 = E_0(\xi), \\ \ell(g(\xi)) &\le (1-\tau)\ell(g_{x_i^{(0)}}) + \tau \ell(g_{x_j^{(1)}}) \\ &\le a_j^{(1)} + \lambda_1 (1-\tau) b \le a_j^{(1)} + \lambda_1 r_1 = E_1(\xi). \end{aligned} \tag{17}$$

At projected targets the same bounds follow directly from (5). The minimizing-index rule is Borel measurable; assigning the finitely many observed values preserves measurability and $|g| \le \max_y |g_y|$. Integrating (17) against optimal empirical-to-$\mathbb{P}_k$ couplings yields, for every $\mathbb{P}_k \in \mathcal{P}_k$,

$$\mathbb{E}_{\mathbb{P}_0} \ell(-g(X[S])) \le \frac{1}{n_0} \sum_i a_i^{(0)} + \rho_0 \lambda_0,$$

$$\mathbb{E}_{\mathbb{P}_1} \ell(g(X[S])) \le \frac{1}{n_1} \sum_j a_j^{(1)} + \rho_1 \lambda_1.$$

Therefore

$$\max_{\gamma} H(\gamma) \le V(S) \le \inf_{(g,a,\lambda)} J(a,\lambda) = \max_{\gamma} H(\gamma).$$

Since $J(\mathbf{1}, 0) = 2$ at $g = 0$, the sublevel $J \le 2$ is nonempty; its feasible points satisfy

$$0 \le a_i^{(k)} \le 2n_k, \qquad 0 \le \lambda_k \le 2/\rho_k.$$

The loss constraints bound both $\ell(g_y)$ and $\ell(-g_y)$; $\ell(t) \to \infty$ as $t \to \infty$ therefore bounds every g_y. The sublevel is compact. Thus the finite minimum and its recovered detector attain $V(S)$, and every optimal solution satisfies

$$\rho_0\lambda_0 + \rho_1\lambda_1 \le J(a,\lambda) = V(S) \le 2. \tag{18}$$

The finite LFD maximum is attained because H is upper semicontinuous on the compact transport set. □

B Proof of Proposition 2.2

Proof. The bounds in Appendix A and $0 \le w_{km} \le \lambda_k$ make the nonempty sublevel $J \le 2$ compact, so $F(z)$ attains its minimum. At $z = \mathbf{1}_S$, (18) permits the lift $w_{km} = \lambda_k z_m$ of a fixed-selection optimum. Conversely,

$$0 \le w_{km} \le \lambda_k z_m \quad \Longrightarrow \quad \sum_m w_{km}|x_i^{(k)}[m] - y[m]| \le \lambda_k \|x_i^{(k)}[S] - y[S]\|_1,$$

which proves (7).

Associate nonnegative multipliers γ, β, μ with the loss constraints, $w_{km} \le \lambda_k$, and the coordinate-capacity constraints, respectively. With g, a free and $\lambda, w \ge 0$, collecting terms in the Lagrangian gives

$$\begin{aligned}\mathcal{L} = & -2\mu^\top z + \sum_{y\in\mathcal{X}} \left[\ell(-g_y)\sum_{i=1}^{n_0}\gamma_{i,y}^{(0)} + \ell(g_y)\sum_{j=1}^{n_1}\gamma_{j,y}^{(1)}\right] \\ & + \sum_{k=0}^{1}\sum_{i=1}^{n_k} a_i^{(k)}\left(\frac{1}{n_k} - \sum_{y\in\mathcal{X}}\gamma_{i,y}^{(k)}\right) + \sum_{k=0}^{1}\lambda_k\left(\rho_k - \sum_m \beta_{km}\right) \\ & + \sum_{k,m} w_{km}\left[\beta_{km} + \rho_k\mu_m - \sum_{i=1}^{n_k}\sum_{y\in\mathcal{X}}\gamma_{i,y}^{(k)}|x_i^{(k)}[m] - y[m]|\right].\end{aligned} \tag{19}$$

Minimization over a is finite precisely when $\sum_y \gamma_{i,y}^{(k)} = 1/n_k$; minimization over g gives $H(\gamma)$. Minimization over $\lambda, w \ge 0$ is finite precisely when

$$\sum_m \beta_{km} \le \rho_k, \qquad \beta_{km} \ge \sum_{i=1}^{n_k}\sum_{y\in\mathcal{X}}\gamma_{i,y}^{(k)}|x_i^{(k)}[m] - y[m]| - \rho_k\mu_m.$$

A nonnegative β satisfying these inequalities exists if and only if

$$\sum_{m=1}^{D}\left(\sum_{i=1}^{n_k}\sum_{y\in\mathcal{X}}\gamma_{i,y}^{(k)}|x_i^{(k)}[m] - y[m]| - \rho_k\mu_m\right)_+ \le \rho_k, \qquad k = 0, 1.$$

The dual feasible set and objective are therefore $\mathcal{D}$ and $H(\gamma) - 2\mu^\top z$.

For every $z \in [0,1]^D$, the choices

$$g = 0, \qquad \lambda = w = 0, \qquad a_i^{(k)} > \ell(0)$$

satisfy all affine constraints and strictly satisfy all loss constraints. Refined Slater duality, with nonnegativity included among the affine constraints, gives equality and dual attainment in (8), including when

$z_m = 0$. Consequently, an optimizer (γ^t, μ^t) at z^t satisfies

$$\begin{aligned} c_t(z) &= H(\gamma^t) - 2(\mu^t)^\top z \le F(z), \\ c_t(z^t) &= H(\gamma^t) - 2(\mu^t)^\top z^t = F(z^t), \\ c_t(z) &= F(z^t) - 2(\mu^t)^\top (z - z^t), \end{aligned}$$

which proves (11). □

References

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